RAJALAKSHMI ENGINEERING COLLEGE

Hasini N 250701237-2025-CSE

V2

Started on	Friday, 25 September 2026, 3:25 PM
State	Finished
Completed on	Friday, 25 September 2026, 3:30 PM
Time taken	4 mins 40 secs
Marks	1.00/1.00
Grade	**4.00** out of 4.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d", &n);
    int a[n];
    for(int i=0; i<n; i++){
        scanf("%d", &a[i]);
    }
    for(int i=0; i<n; i++){
        for(int j=i+1; j<n; j++){
            if(a[i] == a[j]){
                printf("%d", a[i]);
                return 0;
            }
        }
    }
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Hasini N 250701237-2025-CSE

V2

Started on	Friday, 25 September 2026, 3:06 PM
State	Finished
Completed on	Friday, 25 September 2026, 3:15 PM
Time taken	8 mins 19 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 10.00 out of 10.00

Playing with Chessboard:

Ram is given with an n*n chessboard with each cell with a monetary value. Ram stands at the (0,0), that the position of the top left white rook. He is been given a task to reach the bottom right black rook position (n-1, n-1) constrained that he needs to reach the position by traveling the maximum monetary path under the condition that he can only travel one step right or one step down the board. Help ram to achieve it by providing an efficient DP algorithm.

Example:

Input

3

1 2 4

2 3 4

8 7 1

Output:

19

Explanation:

Totally there will be 6 paths among that the optimal is

Optimal path value:1+2+8+7+1=19

Input Format

First Line contains the integer n

The next n lines contain the n*n chessboard values

Output Format

Print Maximum monetary value of the path

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d", &n);
    int a[n][n];
    long long dp[n][n];
    for(int i=0; i<n; i++){
        for(int j=0; j<n; j++){
            scanf("%d", &a[i][j]);
        }
    }
    dp[0][0] = a[0][0];
    for(int j=1; j<n; j++){
        dp[0][j] = dp[0][j-1] + a[0][j];
    }
    for(int i=1; i<n; i++){
        dp[i][0] = dp[i-1][0]+a[i][0];
    }
    for(int  i=1; i<n; i++){
        for(int j=1; j<n; j++){
            if(dp[i-1][j] > dp[i][j-1]){
                dp[i][j] = dp[i-1][j]+a[i][j];
            }
            else{
                dp[i][j]=dp[i][j-1]+a[i][j];
            }
        }
    }
    printf("%lld", dp[n-1][n-1]);
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 2 4 2 3 4 8 7 1	19	19	✔
✔	3 1 3 1 1 5 1 4 2 1	12	12	✔
✔	4 1 1 3 4 1 5 7 8 2 3 4 6 1 6 9 0	28	28	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

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Hasini N 250701237-2025-CSE

V2

Started on	Friday, 25 September 2026, 3:48 PM
State	Finished
Completed on	Friday, 25 September 2026, 3:57 PM
Time taken	8 mins 32 secs
Marks	1.00/1.00
Grade	**30.00** out of 30.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array

2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int T, n,m, i, j;
    scanf("%d", &T);
    while(T--){
        scanf("%d", &n);
        int a[n];
        for(i=0; i<n; i++) scanf("%d", &a[i]);

        scanf("%d", &m);
        int b[m];
        for(i=0; i<m; i++) scanf("%d", &b[i]);

        i=j=0;
        while(i<n && j<m){
            if(a[i] == b[j]){
                printf("%d ", a[i]);
                i++;
                j++;
            }
            else if(a[i] < b[j])
                i++;
```

```
            i++;
        else
            j++;
    }
    printf("\n");
}
return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Hasini N 250701237-2025-CSE

V2

Started on	Friday, 25 September 2026, 4:02 PM
State	Finished
Completed on	Friday, 25 September 2026, 4:19 PM
Time taken	16 mins 45 secs
Marks	1.00/1.00
Grade	**30.00** out of 30.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int main() {
    int t, n1, n2, i, j;
    if (scanf("%d", &t) != 1) return 0;
    while (t--) {
        scanf("%d", &n1);
        int a1[n1];
        for (i = 0; i < n1; i++) scanf("%d", &a1[i]);
        scanf("%d", &n2);
        int a2[n2];
        for (i = 0; i < n2; i++) scanf("%d", &a2[i]);
        i = 0; j = 0;
        while (i < n1 && j < n2) {
            if (a1[i] < a2[j]) i++;
            else if (a1[i] > a2[j]) j++;
            else {
                printf("%d ", a1[i]);
                i++; j++;
            }
        }
        printf("\n");
    }
```

```
    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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V2

Started on	Friday, 25 September 2026, 4:05 PM
State	Finished
Completed on	Friday, 25 September 2026, 4:18 PM
Time taken	13 mins 19 secs
Marks	1.00/1.00
Grade	**4.00** out of 4.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main() {
    int n, k, i = 0, j = 1;
    if (scanf("%d", &n) != 1) return 0;
    int a[n];
    for (int m = 0; m < n; m++) scanf("%d", &a[m]);
    if (scanf("%d", &k) != 1) return 0;
    while (j < n) {
        int diff = a[j] - a[i];
        if (diff == k && i != j) {
            printf("1\n");
            return 0;
        }
        if (diff < k) j++;
        else i++;
        if (i == j) j++;
    }
    printf("0\n");
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔

	Input	Expected	Got	
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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V2

Started on	Friday, 25 September 2026, 4:06 PM
State	Finished
Completed on	Friday, 25 September 2026, 4:19 PM
Time taken	13 mins 10 secs
Marks	1.00/1.00
Grade	**4.00** out of 4.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main() {
    int n, k, i = 0, j = 1;
    if (scanf("%d", &n) != 1) return 0;
    int a[n];
    for (int m = 0; m < n; m++) scanf("%d", &a[m]);
    if (scanf("%d", &k) != 1) return 0;
    while (j < n) {
        int diff = a[j] - a[i];
        if (diff == k && i != j) {
            printf("1\n");
            return 0;
        }
        if (diff < k) j++;
        else i++;
        if (i == j) j++;
    }
    printf("0\n");
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔

	Input	Expected	Got	
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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